

Customer Behavior Analysis

About the Dataset

- **Type:** Retail / E-commerce Dataset
 - **Nature:** Structured (Tabular)
 - **Level:** Customer Transaction Level
 - **Data Mix:**
 - Numerical → Age, Purchase Amount, Previous Purchases
 - Categorical → Category, Payment Method, Season
- 👉 This dataset is used for **customer behaviour analysis + business decision-making**

Domain Background

In the e-commerce industry, companies track every customer interaction:

- What customers buy
- How often do they buy
- How much do they spend
- Whether they respond to discounts

Why This Data Is Important

This data helps businesses:

- Identify high-value customers
- Understand purchase patterns
- Optimize discount strategies
- Improve marketing campaigns
- Increase customer lifetime value (CLV)

Business Problem

How can a retail company understand customer purchasing behaviour to increase revenue,
improve customer retention, and optimize marketing strategies?

Breaking Down the Problem

1. Low Customer Retention

- Customers are not purchasing frequently

Need to understand:

- Who are frequent buyers?
- Who are one-time buyers?

2. Ineffective Marketing Campaigns

- Discounts and promo codes may not be effective

Questions:

- Do discounts increase purchases?
- Which customers respond to promo codes?

3. Product Demand Understanding

- Which category performs best?
- Which products generate the highest revenue?

4. Payment Behaviour Insights

- Which payment methods are preferred?
- Are digital payments increasing?

5. Seasonal Demand Variation

- Does season impact sales?
- Which season drives maximum purchases?

6. Customer Segmentation Problem

Need to identify:

- High-value customers

- Frequent buyers
- Discount-sensitive customers

Data Dictionary

Column Name	Description
Customer ID	Unique identifier for each customer
Age	Age of the customer
Gender	Gender of the customer (Male/Female/Other)
Item Purchased	Specific product purchased
Category	Product category (Clothing, Electronics, etc.)
Purchase Amount	Amount spent by the customer
Location	Customer's geographic location
Size	Product size (S, M, L, etc.)
Colour	Product colour
Season	Season of purchase (Winter, Summer, etc.)
Review Rating	Customer rating for the product
Subscription Status	Whether customer is subscribed (Yes/No), Subscription Program
Shipping Type	Type of shipping selected
Discount Applied	Whether discount was applied
Previous Purchases	Number of past purchases
Payment Method	Mode of payment
Frequency of Purchases	Purchase frequency (Weekly, Monthly, etc.)

EDA- (Exploratory Data Analysis using Python)

1. Data Loading

What I did:

- Imported Pandas
- Loaded dataset using:

```
#importing pandas library
import pandas as pd
```

Why did I do this:

- To bring raw data into a structured DataFrame
- Enables further analysis and transformations

df.head()

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Previous Purchases	Payment Method
0	2701	22	Female	T-shirt	Clothing	68.0	California	XL	Olive	Winter	3.2	No	Standard	No	36.0	Cash
1	521	51	Male	Sunglasses	Accessories	84.0	South Carolina	M	White	Spring	3.9	Yes	Free Shipping	Yes	20.0	Debit Card
2	3157	18	Female	Shirt	Clothing	50.0	Montana	M	Black	Winter	3.1	No	2-Day Shipping	No	18.0	Cash
3	1687	22	Male	Gloves	Accessories	75.0	Illinois	L	Red	Fall	4.2	No	Store Pickup	No	25.0	Cash
4	2929	40	Female	Jewelry	Accessories	80.0	Alabama	L	Yellow	Spring	3.6	No	Store Pickup	No	17.0	Credit Card

2. Initial Data Exploration

What I did:

- Viewed data using:
 - df.head()
 - df.tail()
- Checked dataset summary:
 - df.describe(include='all')
 - df.info()

- Checked missing values:
 - `df.isnull().sum()`

Why did I do this:

- To understand:
 - Data structure
 - Data types
 - Missing values
- This step helps identify data quality issues early

3. Data Consistency Check (Category Correction)

What I did:

- Checked unique values in:
 - Item Purchased
 - Category
- Found inconsistencies between **Item Purchased and Category**
- Created a **mapping dictionary**
- Updated category using:

Why did I do this:

- Categories were inconsistent or incorrect
- Ensures:
 - Accurate grouping
 - Correct business insights

This is a **critical real-world data cleaning step**

4. Handling Missing Values (Size Column)

What you did:

Handled missing values based on category:

- Electronics → "Not Applicable."

- Accessories → "Not Applicable."
- Footwear → "Free Size."
- Clothing → Filled with mode

Why did I do this:

- Different categories require different business logic
- Example:
 - Electronics don't have size → Not Applicable
 - Clothing → size is important → use mode

This shows **domain understanding**

5. Handling Missing Values (Review Rating)

What you did:

- Checked:
 - Mean, median, mode
- Filled missing values using:
- Rounded values

Why did you do it:

- Instead of the global average, the product-level mean is used
- This keeps ratings realistic and meaningful

This is an **advanced imputation technique**

6. Handling Missing Values (Purchase Amount)

What I did: Replaced with the mean value.

Why did I do it?

- Maintains product-level pricing consistency
- Avoids distortion in revenue analysis

7. Handling Missing Values (Previous Purchases)

What you did: Replaced with the 0 (zero) value.

Why did I do it:

- Missing values imply no previous purchase
- Important for:
 - Customer segmentation
 - Loyalty analysis

8. Duplicate Handling

What I did:

- Checked duplicates:
- Identified duplicates using Customer ID
- Removed duplicates:

Why did I do it?

- Ensures:
 - Each customer appears only once
 - No double-counting in analysis

Critical for accurate reporting

9. Column Standardization

What I did:

- Converted column names:

```
df.columns = df.columns.str.replace(" ", "_")
```

```
df.columns = df.columns.str.lower()
```

- Renamed column:

Renaming the Headers ¶

```
df.columns=df.columns.str.replace(" ", "_")  
df.columns=df.columns.str.lower()
```

Why did I do it?

- Makes column names:
 - Clean
 - SQL-friendly
 - Easy to use in Power BI

10. Data Export to PostgreSQL

What I did:

- Installed required libraries:

```
pip install psycopg2-binary sqlalchemy
```

- connected to SQL Server
- Exported data:

Why did you do it?

- To move cleaned data into SQL for further analysis
- Enables:
 - Query-based insights
 - Integration with BI tools.

Data Analysis using PostgreSQL

1. Which category generates the highest revenue?

	category text	total_revenue numeric
1	Electronics	486366.96
2	Accessories	194866.87
3	Clothing	183188.39
4	Footwear	111236.20
5	Outerwear	18524.00

2. Are discounts actually increasing purchase value?

	discount_applied text	total_revenue numeric	avg_purchase numeric
1	No	502241.64	182.24
2	Yes	491940.78	219.22

3. What is the total revenue generated by male and female customers?

	gender text	highest_revenue numeric
1	Male	390964.53
2	Female	352060.29
3	Other	251157.60

4. Which customer used a discount but still spent more than the average purchase amount?

	customer_id bigint 	purchase_amount numeric 	discount_applied text 
1	4568	276.34	Yes
2	4256	734.48	Yes
3	4289	1282.44	Yes
4	4576	831.34	Yes
5	4319	685.23	Yes
6	4708	735.77	Yes
7	4584	685.23	Yes
8	4607	557.24	Yes
9	4851	685.23	Yes
10	4845	692.11	Yes

5. Which are the top/bottom 5 products with the highest average review rating?

	item_purchased text 	avg_review_rating numeric 
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.79

	item_purchased text	avg_review_rating numeric
1	Phone	2.94
2	Headphones	3.08
3	Watch	3.10
4	Bag	3.17
5	Laptop	3.19

6. Average purchase: Standard vs Express shipping

	shipping_type text	order_placed bigint	avg_purchase numeric	revenue numeric
1	Express	1199	358.12	429380.98
2	Standard	1201	340.57	409020.45

7. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	No	3421	165.72	566937.02
2	Yes	1579	270.58	427245.40

8. Top 5 products with highest discount usage %

	item_purchased text	total_number_of_times_sold bigint	number_of_times_sold_when_disocunt_applied bigint	discount_percent numeric
1	Laptop	143	74	51.75
2	Phone	171	88	51.46
3	Headphones	166	85	51.20
4	Hat	154	77	50.00
5	Watch	157	78	49.68

9. Segment customers into new, returning, and loyal based on their total number of previous purchases, and show the count of each segment.

	customer_segment text	customer_count bigint
1	Loyal Customers	3085
2	New Customer	561
3	Returning Customer	1354

10. What are the top 3 most purchased products within each category?

	category text	item_purchased text	most_purchased bigint	rnk bigint
1	Accessories	Jewelry	171	1
2	Accessories	Bag	164	2
3	Accessories	Belt	161	3
4	Accessories	Sunglasses	161	3
5	Clothing	Shirt	315	1
6	Clothing	Pants	171	2
7	Clothing	Blouse	171	2
8	Electronics	Phone	171	1
9	Electronics	Headphones	166	2
10	Electronics	Watch	157	3
11	Footwear	Shoes	303	1
12	Footwear	Sandals	160	2
13	Footwear	Sneakers	145	3
14	Outerwear	Jacket	163	1
15	Outerwear	Coat	161	2

11. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

	customer_type text	subscription_status text	customer_count bigint	percents numeric
1	Normal Buyers	No	658	62.85
2	Normal Buyers	Yes	389	37.15
3	Repeat Buyers	No	2763	69.90
4	Repeat Buyers	Yes	1190	30.10

12. What is the revenue contribution of each age group?

	age_group text	total_revenue numeric
1	51+	399345.32
2	35-50	266706.70
3	26-35	177421.93
4	18-25	150708.48

Power BI Dashboard

